IFF Duration Predictor

- Predict Duration
- Optimize Ingredients
- Model Dashboard

Powered by XGBoost

Compare XGBoost, LightGBM, and Random Forest performance metrics

Best Model	Best MAE	Best R^2
LightGBM	**0.576h**	**0.9905**
Selected by lowest MAE	Mean Absolute Error	Variance Explained

Model Comparison (Regression)

MODEL	MAE (HOURS)	RMSE (HOURS)	R^2	MAPE (%)
XGBoost	0.608	0.845	0.99	10.07%
LightGBM BEST	0.576	0.823	0.9905	10.73%
RandomForest	0.926	1.361	0.9741	14.4%

MAE Comparison (lower is better)

XGBoost	0.608h
LightGBM	0.576h

R^2 Comparison (higher is better)

XGBoost	0.99
LightGBM	0.9905

RandomForest 0.926h

RandomForest 0.9741

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Understanding the Metrics

MAE (Mean Absolute Error)

Average prediction error in hours. A MAE of 1.5 means predictions are off by 1.5 hours on average.

RMSE (Root Mean Squared Error)

Like MAE but penalizes large errors more. Lower is better.

R^2 (Coefficient of Determination)

Proportion of variance explained. 1.0 = perfect, 0.0 = baseline. Values above 0.85 indicate strong predictions.

MAPE (Mean Absolute % Error)

Prediction error as a percentage. Useful for comparing across different duration ranges.

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